8.7 Contrasts

In the one-way model the individual effects α_k are not estimable, but their differences are. The general form of such a difference is a contrast. Contrasts are the natural targets of inference about a factor. They are estimable, they do not depend on the side condition or coding, and they are what a scientific question about a factor usually asks: does treatment beat control, is there a trend across doses, do two groups of levels differ? This section characterizes them, shows how an orthogonal set of contrasts splits the factor's sum of squares, and constructs the Helmert and polynomial families.

8.7.1 Definition and estimability

Throughout, the one-way model is $\mathrm{E}(y_{kj}) = \mu + \alpha_k$ for observation j at level k, with $n_k \geq 1$ observations at level $k = 1, \ldots, g$, level means $\mu_k = \mu + \alpha_k$ and sample means $\bar{y}_k$.

Definition 8.7.1 — Contrast

In a factor model, a **contrast** in the effects of a factor is a linear function $\sum_k c_k \alpha_k$ whose coefficients satisfy $\sum_k c_k = 0$, not all zero. In the one-way model it equals $\sum_k c_k \mu_k$, and we call either form a contrast in the level means. Two contrasts are **orthogonal** (for the given design) if $\sum_k c_k d_k / n_k = 0$.

The two forms agree because $\sum_k c_k \mu_k = \mu \sum_k c_k + \sum_k c_k \alpha_k$. This is where the condition $\sum_k c_k = 0$ comes from: it makes the unidentified μ drop out. Simple differences $\alpha_k - \alpha_l$, comparisons of one level with the average of others such as $\alpha_1 - \frac{1}{2}(\alpha_2 + \alpha_3)$, and trend coefficients are all contrasts.

Theorem 8.7.1 — Estimable functions in the one-way model

In the one-way model with every $n_k \geq 1$, the function $\lambda_0 \mu + \sum_k \lambda_k \alpha_k$ is estimable iff $\lambda_0 = \sum_k \lambda_k$. In particular, a linear function of the effects alone, $\sum_k c_k \alpha_k$, is estimable iff it is a contrast (or zero). For a contrast, the least squares estimator and its variance under $\mathrm{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ are

$$\hat{\psi} = \sum_k c_k \bar{y}_k, \qquad \mathrm{Var}(\hat{\psi}) = \sigma^2 \sum_k \frac{c_k^2}{n_k}. \tag{8.7.1}$$

Proof. The null space of $\mathbf{X} = [\mathbf{1}, \mathbf{Z}]$ is spanned by $\mathbf{v} = (1, -1, \ldots, -1)^\top$ (Example (A rank-deficient layout)). By Theorem 8.3.1(c), the function is estimable iff $\lambda_0 - \sum_k \lambda_k = 0$. For an estimable function, $\lambda_0 \mu + \sum_k \lambda_k \alpha_k = \sum_k \lambda_k (\mu + \alpha_k)$. Every least squares solution has $\hat{\mu} + \hat{\alpha}_k = \bar{y}_k$, since the fitted values are the level means (Example (The one-way layout)). So the estimate is $\sum_k \lambda_k \bar{y}_k$. The level means are uncorrelated with variances σ^2 / n_k, by Theorem 2.2.1, which gives the variance.

■

The estimator in Eq. 8.7.1 is the same whatever coding or side condition was used to fit the model. Proposition 8.3.2 guarantees this, and it is the reason contrasts, rather than coefficients, should be reported.

8.7.2 Sums of squares and orthogonal contrasts

Each contrast defines a direction in the observation space, and its estimate is the component of $\mathbf{y}$ along that direction. With $\mathbf{D} = \operatorname{diag}(n_1, \dots, n_g)$ and $\mathbf{Z}$ the indicator matrix, put

$$\mathbf{u_c} = \mathbf{Z}\mathbf{D}^{-1}\mathbf{c}, \tag{8.7.2}$$

the vector whose entries equal c_k/n_k for the observations at level k.

Proposition 8.7.2 — Contrasts as projections

Let $\mathbf{c}$ and $\mathbf{d}$ be contrasts, and write $\hat{\psi}_{\mathbf{c}} = \mathbf{c}^\top \hat{\boldsymbol{\mu}}$. Then:

(a) $\hat{\psi}_{\mathbf{c}} = \mathbf{u}_{\mathbf{c}}^\top \mathbf{y}$, and $\mathbf{u_c}$ is a nonzero vector in $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$;

(b) the **sum of squares for the contrast**,

$$\mathrm{SS}(\mathbf{c}) = \frac{\hat{\psi}_{\mathbf{c}}^2}{\sum_k c_k^2/n_k},$$

equals $\|\mathbf{P}\mathbf{y}\|^2$, where $\mathbf{P}$ is the orthogonal projection onto the line spanned by $\mathbf{u_c}$;

(c) $\operatorname{Cov}(\hat{\psi}_{\mathbf{c}}, \hat{\psi}_{\mathbf{d}}) = \sigma^2 \sum_k c_k d_k / n_k = \sigma^2 \mathbf{u}_{\mathbf{c}}^\top \mathbf{u_d}$ when $\operatorname{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, so orthogonal contrasts have uncorrelated estimates and orthogonal vectors $\mathbf{u}$;

(d) if $\mathbf{c}_1, \dots, \mathbf{c}_{g-1}$ are pairwise orthogonal contrasts, then

$$\sum_{j=1}^{g-1} \mathrm{SS}(\mathbf{c}_j) = \sum_{k=1}^{g} n_k (\bar{y}_k - \bar{y})^2,$$

the sum of squares between levels.

Proof.

(a) $\hat{\boldsymbol{\mu}} = \mathbf{D}^{-1}\mathbf{Z}^\top \mathbf{y}$ (Eq. 8.5.1), so $\mathbf{c}^\top \hat{\boldsymbol{\mu}} = (\mathbf{Z}\mathbf{D}^{-1}\mathbf{c})^\top \mathbf{y}$. Clearly $\mathbf{u_c} \in \mathcal{C}(\mathbf{Z})$, and $\mathbf{1}^\top \mathbf{u_c} = \mathbf{1}^\top \mathbf{Z}\mathbf{D}^{-1}\mathbf{c} = (n_1, \dots, n_g)\mathbf{D}^{-1}\mathbf{c} = \sum_k c_k = 0$. It is nonzero because $\mathbf{Z}\mathbf{D}^{-1}$ has full column rank and $\mathbf{c} \neq \mathbf{0}$.

(b) By Example (Projections onto a line), $\|\mathbf{P}\mathbf{y}\|^2 = (\mathbf{u}_{\mathbf{c}}^\top \mathbf{y})^2 / \|\mathbf{u_c}\|^2$, and $\|\mathbf{u_c}\|^2 = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{Z}^\top \mathbf{Z}\mathbf{D}^{-1}\mathbf{c} = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{c} = \sum_k c_k^2/n_k$.

(c) Theorem 2.2.1 gives $\operatorname{Cov}(\mathbf{u}_{\mathbf{c}}^\top \mathbf{Y}, \mathbf{u}_{\mathbf{d}}^\top \mathbf{Y}) = \sigma^2 \mathbf{u}_{\mathbf{c}}^\top \mathbf{u_d}$, and the same computation as in (b) gives $\mathbf{u}_{\mathbf{c}}^\top \mathbf{u_d} = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{d}$.

(d) The space $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$ has dimension $g - 1$ (Theorem 6.5.1(c), with $\mathcal{C}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{Z})$). By (a) and (c) the vectors $\mathbf{u}_{\mathbf{c}_1}, \dots, \mathbf{u}_{\mathbf{c}_{g-1}}$ are $g - 1$ nonzero, mutually orthogonal vectors in it, so they form an orthogonal basis. By Theorem 6.3.8 the sum of the projections onto the lines they span is the projection onto that space, which is $\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top$ (Theorem 6.5.1(b)). Applying it to $\mathbf{y}$ and using Pythagoras gives

$\sum_j \mathrm{SS}(\mathbf{c}_j) = \|(\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}\|^2$. The vector $(\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}$ has entry $\bar{y}_k - \bar{y}$ at every observation of level k.

■

So a complete set of orthogonal contrasts decomposes the between-level sum of squares into $g-1$ one-dimensional pieces. Under normality the pieces are independent, each is σ^2 times a noncentral χ^2 with one degree of freedom, and each is independent of the residual sum of squares (Theorem 4.4.4). Each contrast can therefore be tested on its own. Chapter 9 places this decomposition in the general theory of orthogonal sums of squares, and Chapter 15 uses it for the one-way analysis of variance (Theorem 15.4.1). In a *balanced* design, with all n_k equal, orthogonality reduces to the ordinary $\sum_k c_k d_k = 0$. In an unbalanced design, contrasts that look orthogonal on paper need not be orthogonal for the data.

8.7.3 Helmert contrasts

The Helmert matrix of Example (Helmert matrices) has rows proportional to

$$\mathbf{h}_k = (\underbrace{1, \dots, 1}_{k-1}, -(k-1), 0, \dots, 0), \qquad k = 2, \dots, g.$$

The contrast $\mathbf{h}_k^\top \boldsymbol{\mu}$ compares level k with the average of the levels before it, since it is $(k-1)$ times $\frac{1}{k-1}\sum_{l<k}\mu_l - \mu_k$. The $\mathbf{h}_k$ are mutually orthogonal in the ordinary sense, so in a balanced design they are orthogonal contrasts. They suit factors whose levels are added in a meaningful order, such as successive refinements of a treatment, where each new level is compared with everything before it.

8.7.4 Polynomial contrasts

When the levels of a factor are ordered and equally spaced, such as doses $0, 10, 20, 30$ or the seven points of a party scale, the natural questions are about the *shape* of the sequence of means. Is there a linear trend? Curvature? Give the levels scores $s_k = k$. Apply Gram–Schmidt (Proposition 6.2.2) to the vectors $\mathbf{1}, \mathbf{s}, \mathbf{s}^2, \dots, \mathbf{s}^{g-1}$ in $\mathbb{R}^g$, where $\mathbf{s}^j$ has entries s_k^j. This gives orthogonal vectors $\mathbf{p}_0, \dots, \mathbf{p}_{g-1}$, and $\mathbf{p}_j$ is a polynomial of degree j in the score. For $j \geq 1$, $\mathbf{p}_j \perp \mathbf{1}$, so $\mathbf{p}_j$ is a contrast. Rescaled to integers, the first three for $g = 7$ are

$$\text{linear } (-3, -2, -1, 0, 1, 2, 3), \quad \text{quadratic } (5, 0, -3, -4, -3, 0, 5), \quad \text{cubic } (-1, 1, 1, 0, -1, -1, 1).$$

By construction $\mathbf{p}_j$ is orthogonal to every polynomial of degree less than j. So if the level means follow a polynomial of degree d in the score, every contrast of degree above d is zero. The linear contrast measures the trend, the quadratic the curvature beyond it, and so on. This is the reason for their use.

With unequal group sizes the classical integer contrasts are no longer orthogonal for the design. The fix follows Proposition 8.7.2(c). Orthogonality is needed in the inner product $\sum_k c_k d_k / n_k$, which suggests setting $\mathbf{c}_j = \mathbf{D}\mathbf{p}_j$, where now the $\mathbf{p}_j$ are orthogonal polynomials in the weighted inner product $\sum_k n_k p_k q_k$. Then $\sum_k c_{jk} = \sum_k n_k p_{jk} = 0$, and $\sum_k c_{jk} c_{lk}/n_k = \sum_k n_k p_{jk} p_{lk} = 0$ for $j \neq l$. In observation space, $\mathbf{u}_{\mathbf{c}_j} = \mathbf{Z}\mathbf{p}_j$ is the polynomial of degree j in each observation's score, orthogonalized over the observations. The weighted linear contrast gives exactly the regression sum of squares of $\mathbf{y}$ on the score (Exercise (C1)).

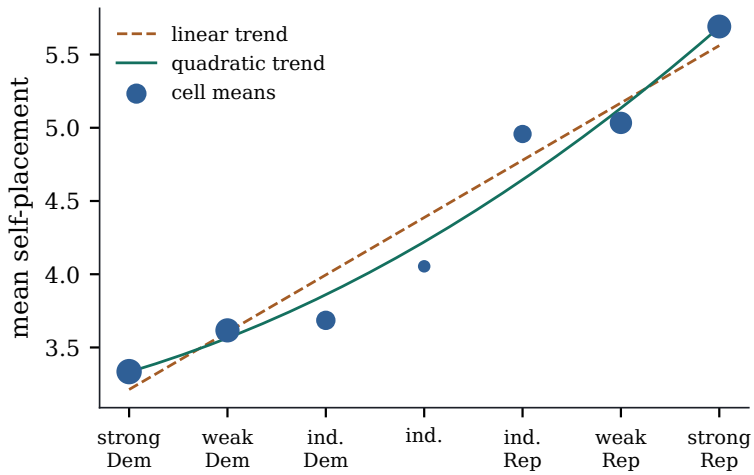

Figure 8.7.1: Mean ideological self-placement by party identification (point area proportional to group size), with the weighted least squares linear and quadratic trends in the score. The linear contrast accounts for almost all of the variation between groups.

Example. (Trend in ideology across party identification)

Return to the seven party-identification groups of Example (Party identification and ideology). The between-group sum of squares is 772.87. The classical linear contrast has estimate $\hat{\psi} = \sum_k (k-4)\bar{y}_k = 11.175$, variance $0.1652\,\sigma^2$ by Eq. 8.7.1, and sum of squares 755.85, which is 97.8 percent of the between-group sum of squares. The quadratic and cubic contrasts have sums of squares 10.15 and 2.58. Ideology rises almost linearly across the seven-point party scale, with some upward curvature (Figure 8.7.1).

The group sizes range from 37 to 200, so the six classical contrasts are not orthogonal for this design. Their sums of squares add to 784.02, which exceeds the between-group total. Part of the variation is counted twice. The weighted orthogonal polynomial contrasts decompose the total exactly: 746.57 for the linear trend and 26.30 for the five higher-degree contrasts together.

```
import numpy as np
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].to_numpy()
level = anes["PID"].to_numpy().astype(int)        # 0 = strong Dem. ... 6 = strong
    Rep.
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
```

Run this code online

```
def orthogonal_polynomials(scores, weights):
    """Columns 1..g-1: polynomials in the scores, orthogonal in sum_k w_k u_k v_k.
    """
    V = np.vander(scores, len(scores), increasing=True)       # 1, s, s^2, ...
    W = np.sqrt(weights)[:, None]
    Q, _ = np.linalg.qr(W * V)                                # weighted Gram-Schmidt
```

```
    P = Q / W                                                # back to the original
    scale
    P = P / np.abs(P).max(axis=0)                            # cosmetic rescaling
    return P[:, 1:]                                          # drop the constant
    column

def contrast_ss(c, means, n_k):
    """Estimate c'mu-hat and its sum of squares (c'mu-hat)^2 / sum(c_k^2 / n_k)."""
    psi = c @ means
    return psi, psi ** 2 / np.sum(c ** 2 / n_k)

scores = np.arange(g, dtype=float)
C_unw = orthogonal_polynomials(scores, np.ones(g))           # the classical table (
    equal n)
C_wtd = n_k[:, None] * orthogonal_polynomials(scores, n_k)   # c_k = n_k p(k)
ss_between = np.sum(n_k * (means - y.mean()) ** 2)
for name, C in [("unweighted", C_unw), ("weighted", C_wtd)]:
    ss = [contrast_ss(C[:, j], means, n_k)[1] for j in range(g - 1)]
    print(f"{name:10s} SS by degree: {np.round(ss, 2)}  total {sum(ss):.2f}",
          f"(between groups {ss_between:.2f})")
```

Run this code online

8.7.5 Coding matrices are not contrast matrices

Section 8.5 coded a factor by an intercept and the columns $\mathbf{ZC}$. When the columns of $\mathbf{C}$ are contrasts, as for Helmert or polynomial coding, it is tempting to read the coefficient of column j as the estimate of the contrast $\mathbf{c}_j^\top \boldsymbol{\mu}$. That reading is wrong in general. By Proposition 8.5.1(b) the coefficients are the rows of $\mathbf{K}^{-1}$, with $\mathbf{K} = [\mathbf{1}, \mathbf{C}]$, applied to $\hat{\boldsymbol{\mu}}$. The rows of $\mathbf{K}^{-1}$ and the columns of $\mathbf{K}$ are related simply only when the columns are orthogonal.

Proposition 8.7.3 — Orthogonal codings

If the columns of $\mathbf{K} = [\mathbf{1}_g, \mathbf{c}_1, \ldots, \mathbf{c}_{g-1}]$ are mutually orthogonal, then the coefficients of the coded model are

$$\hat{\gamma}_0 = \frac{1}{g}\sum_k \bar{y}_k, \qquad \hat{\gamma}_j = \frac{\mathbf{c}_j^\top \hat{\boldsymbol{\mu}}}{\mathbf{c}_j^\top \mathbf{c}_j}, \quad j = 1, \ldots, g-1,$$

whatever the group sizes.

Proof. With orthogonal columns, $\mathbf{K}^\top\mathbf{K} = \operatorname{diag}(g, \mathbf{c}_1^\top\mathbf{c}_1, \ldots, \mathbf{c}_{g-1}^\top\mathbf{c}_{g-1})$, so $\mathbf{K}^{-1} = (\mathbf{K}^\top\mathbf{K})^{-1}\mathbf{K}^\top$. Apply Proposition 8.5.1(b). ∎

The orthogonality required here is the plain one, $\mathbf{c}_j^\top \mathbf{c}_l = 0$, not the design orthogonality of Definition 8.7.1, because $\hat{\boldsymbol{\gamma}} = \mathbf{K}^{-1}\hat{\boldsymbol{\mu}}$ is pure algebra and does not involve the n_k. Helmert coding, in the usual software convention, uses the columns $\mathbf{c}_j = -\mathbf{h}_{j+1}$, which compare level $j+1$ with the levels before it, and $\mathbf{c}_j^\top \mathbf{c}_j = j(j+1)$. In the party data the first three Helmert coefficients are 0.1408, 0.0698 and 0.1271, while the contrasts themselves are 0.2817, 0.4187 and

1.5253, larger by the factors 2, 6 and 12. For a non-orthogonal coding there is no such shortcut. Reference coding has $\mathbf{c}_j = \mathbf{e}_{j+1}$, yet its coefficients are $\mu_{j+1} - \mu_1$, not μ_{j+1}.

```
def helmert(g):
    """Coding matrix whose column k compares level k+1 with the mean of levels 1..k.
    """
    C = np.zeros((g, g - 1))
    for k in range(1, g):
        C[:k, k - 1] = -1.0
        C[k, k - 1] = k
    return C

H = helmert(g)
K = np.column_stack([np.ones(g), H])
gamma = np.linalg.solve(K, means)                    # Helmert-coded coefficients
psi = H.T @ means                                    # the contrasts c_k' mu-hat
print("coefficients:", np.round(gamma[1:], 4))
print("contrasts:   ", np.round(psi, 4))
print("ratio:       ", np.round(psi / gamma[1:], 1))    # = c_k' c_k = k(k+1)
```

Run this code online

> *Warning.* A coefficient from a contrast-coded regression estimates a contrast only up to a scale factor, and only when the coding columns are orthogonal. To test or report a contrast, compute $\mathbf{c}^\top \hat{\boldsymbol{\mu}}$ and its variance Eq. 8.7.1 directly, or read it from the coded fit using $\mathbf{K}^{-1}$. Do not read it off the coefficient table.

8.7.6 Contrasts in the two-way additive model

In the additive model $\mu + \alpha_i + \beta_j$ of Section 8.6, a **row contrast** is $\sum_i c_i \alpha_i$ with $\sum_i c_i = 0$. It is a combination of the differences $\alpha_i - \alpha_1$, so in a connected design it is estimable (Theorem 8.6.1). In a balanced design, with $m \geq 1$ observations in every one of the ab cells, the least squares fit is $\bar{y}_{i\cdot\cdot} + \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdot\cdot\cdot}$. This is shown in Example (Additive two-factor model, one observation per cell) for $m = 1$, and the averaging argument there is unchanged for $m > 1$. Hence

$$\widehat{\textstyle\sum_i c_i \alpha_i} = \sum_i c_i \big(\bar{y}_{i\cdot\cdot} + \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdot\cdot\cdot}\big) = \sum_i c_i \bar{y}_{i\cdot\cdot}, \qquad \mathrm{Var} = \frac{\sigma^2}{bm} \sum_i c_i^2,$$

since the terms not depending on i are multiplied by $\sum_i c_i = 0$. The row contrast is estimated from the row means alone, as if factor B were absent. In an unbalanced design this is false. The row means are then contaminated by the column effects (Exercise (B2)), and the least squares estimate has to adjust for them. That adjustment is the subject of Chapter 17. Interaction contrasts, and their estimability when cells are empty, were treated in Proposition 8.6.2.

8.7.7 Exercises

8.7.7.1 A. Check your understanding

> **Exercise A1.**
>
> Find the linear, quadratic and cubic orthogonal polynomial contrasts for $g = 4$ equally spaced levels, scaled to integers. Check that they are mutually orthogonal and that the

quadratic contrast vanishes when $\mu_k = a + bk$.

Exercise A2.

Show that the Helmert contrasts $\mathbf{h}_2, \ldots, \mathbf{h}_g$ are orthogonal for the design iff $n_1 = n_2 = \cdots = n_{g-1}$, whatever n_g is. *Hint:* compute $\sum_k h_{jk} h_{lk}/n_k$ for $j < l$.

8.7.7.2 B. Practice

Exercise B1.

Show that for every contrast $\mathbf{c}$, $\mathrm{SS}(\mathbf{c}) \le \sum_k n_k(\bar{y}_k - \bar{y})^2$, with equality when $c_k = n_k(\bar{y}_k - \bar{y})$. This fact is the basis of Scheffé's method in Chapter 13 (Theorem 13.2.2).

Solution. By Proposition 8.7.2(b), $\mathrm{SS}(\mathbf{c}) = \|\mathbf{P}\mathbf{y}\|^2$ with $\mathbf{P}$ projecting onto a line inside $\mathcal{S} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$. Let $\mathbf{P}_\mathcal{S} = \mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top$. Because the line lies in $\mathcal{S}$, $\mathbf{P}\mathbf{y} = \mathbf{P}\mathbf{P}_\mathcal{S}\mathbf{y}$ (Theorem 6.5.1(a)), and so $\|\mathbf{P}\mathbf{y}\|^2 \le \|\mathbf{P}_\mathcal{S}\mathbf{y}\|^2 = \sum_k n_k(\bar{y}_k - \bar{y})^2$. Equality holds when the line contains $\mathbf{P}_\mathcal{S}\mathbf{y}$, whose entries are $\bar{y}_k - \bar{y}$ at level k. That vector is $\mathbf{u}_\mathbf{c}$ for $c_k = n_k(\bar{y}_k - \bar{y})$, and this $\mathbf{c}$ is a contrast because $\sum_k n_k(\bar{y}_k - \bar{y}) = 0$.

Exercise B2.

In the additive two-way model with n_{ij} observations in cell (i, j), show that $\mathrm{E}(\bar{y}_{i\cdot\cdot}) = \mu + \alpha_i + \sum_j n_{ij}\beta_j/n_{i\cdot}$. Deduce that $\sum_i c_i \bar{y}_{i\cdot\cdot}$ is unbiased for every row contrast and every $\boldsymbol{\beta}$ iff the proportions $n_{ij}/n_{i\cdot}$ do not depend on i, that is, iff the cell counts are *proportional*, $n_{ij} = n_{i\cdot}n_{\cdot j}/n$.

8.7.7.3 C. Going deeper

Exercise C1.

Let $\mathbf{x}$ be the n-vector giving each observation's score s_k. Show that the sum of squares of the weighted linear contrast $\mathbf{c}_1 = \mathbf{D}\mathbf{p}_1$ equals the regression sum of squares $\|(\mathbf{M}_1 - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}\|^2$ of the simple regression of $\mathbf{y}$ on $\mathbf{x}$ with intercept, where $\mathbf{M}_1$ projects onto $\mathcal{C}([\mathbf{1}, \mathbf{x}])$. Conclude that the remaining weighted contrasts together measure the lack of fit of the straight line to the level means.

Solution. In the weighted inner product, $\mathbf{p}_1$ is $\mathbf{s}$ minus its weighted mean, $\mathbf{s} - \bar{s}_w\mathbf{1}$ with $\bar{s}_w = \sum_k n_k s_k/n$, up to scale. Then $\mathbf{u}_{\mathbf{c}_1} = \mathbf{Z}\mathbf{D}^{-1}\mathbf{D}\mathbf{p}_1 = \mathbf{Z}\mathbf{p}_1$, whose entries are $s_k - \bar{s}_w$. That is $\mathbf{x} - \bar{x}\mathbf{1}$, the centred score vector. By Proposition 8.7.2(b), $\mathrm{SS}(\mathbf{c}_1)$ is the squared length of the projection of $\mathbf{y}$ onto the line spanned by $\mathbf{x} - \bar{x}\mathbf{1}$. By Example (Intercept inside simple regression) this line is $\mathcal{C}([\mathbf{1}, \mathbf{x}]) \cap \mathbf{1}^\perp$, and the projection onto it is $\mathbf{M}_1 - n^{-1}\mathbf{1}\mathbf{1}^\top$. The remaining weighted contrasts span the orthogonal complement of that line in $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$, which is $\mathcal{C}(\mathbf{Z}) \cap \mathcal{C}([\mathbf{1}, \mathbf{x}])^\perp$. Their total sum of squares is $\|(\mathbf{M} - \mathbf{M}_1)\mathbf{y}\|^2$, the reduction in residual sum of squares from the straight-line model to the model with a separate mean for each level.